Response to Review — CSME 2026 paper

A pvlib-Based Computational Model of a Multi-Inverter Rooftop Photovoltaic Plant Driven by Public Weather Data: Sensitivity Analysis and Field Validation

C. N. Govino, W.-R. Yang — revision of 30 October 2026 (6 pages)

Summary of changes

Two new figures (plant/measurement layout; three-phase waveforms), two new tables (structure vs efficiency; error metrics), a converged Sobol analysis with confidence intervals, a complete lower-bound stack, an aliasing test and a long-term check against the inverter energy counters. Two conclusions changed: the budget can reach the clear-day measurement, but only at an implausible joint extreme; and over 3.5 years the counters agree with the model within 3%, so the clear-day shortfall is not a permanent plant loss. To keep six pages, the model-chain diagram, the monthly ERA5 chart, the hourly-ratio chart, the per-record chart and the single-change table were removed; their key numbers remain in the text.

Reviewer comment	Response and change	Where
1. Empirical validation is insufficient (about three months of data and a few short records).	Agreed; the field data remain limited and this is now stated as limitation (1). The raw Hioki recording did not run continuously, so no further measured days exist. Added a long-term check that needs no new measurement: the lifetime energy counters of the two 50 kW inverters (1280 days since commissioning). Compared with the model for the same calendar period, the counters are 0.8% and 2.7% higher (1411 and 1437 vs 1399 kWh/kWp/yr). The counters read about 3.0% above the recorder (checked on 18 Oct 2026). Conclusion changed: the 19% clear-day shortfall is not a permanent loss; the model is suitable for annual energy but not for hourly prediction on clear days. Suggested future data: monthly kWh of the Taipower sale meter and continuous logging.	Abstract; 4.5; 5; 6
2. Irradiance from a station 21 km away; no on-site measurement, so discrepancies cannot be attributed to plant losses.	Agreed. No pyranometer exists on the roof, so this cannot be fixed with present data. The paper now says explicitly that the model error cannot be split into an irradiance part and a plant part, and names a plane-of-array pyranometer or reference cell as the most important next step. Mitigation already in the paper: only days with station clear-sky fraction > 0.9 are used for the bias; the hour-by-hour pattern and the February/October contrast argue against a constant station bias.	2.2; 4.6; 5 (2)
3. Measurement layout is not shown; thinning 2 ms → 10 ms must be validated for aliasing of the active power.	New Fig. 1: roof blocks, strings per MPPT, the three inverters, AC panel, recorder position and grid connection. New Fig. 2 (as sketched by the supervisor): three phase voltages of the bus, 120° apart, with rms and peak values; Eq. (1) $P = \sqrt{3}\cdot V_{LL}\cdot I_L\cdot\cos\varphi$ and Eq. (2) two-wattmeter power. Text explains why $\Sigma\, v_{LL}\cdot i$ would give 1.5 P. Aliasing: explained why the mean power is preserved (samples at five equally spaced phases; whole-cycle intervals avoided) and tested with 400 synthetic minutes built from the measured unbalance, displacement and harmonics. Max error 0.06% (10 ms) and 0.05% (20 ms); 0.12% and 0.12% even at exactly 60 Hz; 200 ms would err by up to 12%. Note: the raw 2 ms files are not kept; if any are found, the same test can be repeated on real data with src/rev_aliasing.py adapted.	Fig. 1; Fig. 2; 2.3

Reviewer comment	Response and change	Where
4. Sobol ranking depends on ranges and distributions; show convergence, convergence testing and confidence intervals.	Table 2 now gives the range, distribution and basis of every factor and states that the ranking depends on them (the ±10% GHI case shows the shift). Convergence: base samples raised from 1,024 to 4,096 (45,056 runs) plus five extra scrambling seeds. Total indices change by at most 0.003 (±5%) and 0.003 (±10%); seed-to-seed SD ≤ 0.002; top-five ranking stable from 64 samples. Table 2 reports S_T with 95% bootstrap confidence intervals; the last four factors have indices below 0.005 and are not ranked.	3.1; 3.2; Table 2
5. Distinguish differences in model structure from differences in assumed rated efficiency.	New Section 3.3 and Table 3. Same η: constant vs PVWatts differ by 0.36–0.37%; same peak efficiency: 0.6%; changing η from 0.985 to 0.96 within PVWatts: −2.6%. So the inverter effect in the budget is an efficiency assumption, not a structural one. On the clear October days the two structures with equal η differ by < 0.1%.	3.3; Table 3; Abstract; 6
6. Clarify the 'stacked lower-bound assumption' and why options that would give lower output were excluded.	The reviewer is right: the earlier stack left out isotropic transposition (−3.9%) and Boland decomposition (−1.0%). Both are now included (Fig. 6; single effects given in the text). With every lower end the model reaches 82.0% of baseline, i.e. the measurement (81%) is reached. This changed the conclusion. The earlier sentence 'the remainder cannot be explained by the nine factors' was removed. New text: the budget reaches the measurement only at its joint extreme, which is implausible because (i) the isotropic model underestimates tilted irradiance on clear days [27], (ii) all lower ends together give −13.6% annually, far below the 5th percentile (−5.5%), and (iii) the same setting underestimates the clear day of 10 Feb by about 6%.	4.4; Fig. 5; Fig. 6
7. High correlation but no accurate prediction; systematic overestimate of about 19% on clear days.	Agreed and stated directly: correlation shows only the shape; the level is wrong. The ratio is now defined explicitly as $R = \Sigma P_{actual} / \Sigma P_{model}$ (Eq. 4). Table 4 now gives MBE, nMBE, RMSE, nRMSE and r for all five clear-sky records (nMBE +23% and +24%, nRMSE 27% and 31% in October). A leave-one-day-out test shows that a single correction factor does not transfer (factor from 17 Oct leaves −8.6% on 10 Feb and −19.7% on 7 Aug). The paper no longer claims predictive accuracy; it claims agreement in annual energy (counters) and a clear-day bias at hourly scale.	4.2 (Eq. 4); 4.3; Table 4; 5; 6

Code for every new number: src/rev_sobol_convergence.py, src/rev_analysis.py, src/rev_aliasing.py, src/rev_fig_*.py; collected in paper_rev/numbers.json.